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1. INTRODUCTION

1.1 Background of the Study

The COVID-19 pandemic is the most significant economic disruption the world has seen since the Second World War. Beyond its public health impact, the pandemic also had devastating effects on international trade, global supply chains, financial markets and business operations across countries. Restrictions on mobility, lockdowns, transportation bottlenecks, labour shortages and production network disruptions created unparalleled challenges for firms and economies around the world.

The effects were particularly felt in global supply chains, as countries that exported and imported goods were experiencing shortages, delays and fluctuations in import/export. The interconnectedness of modern trade networks amplified the impact of disruptions in key supplier countries, making economies more vulnerable to shocks.

In India, the pandemic hit businesses in all sectors, especially Micro, Small and Medium Enterprises (MSMEs) badly. MSMEs provide significant employment, industrial production, exports and economic development. However, their relatively limited financial resources and market dependency made them particularly vulnerable during the pandemic. To counter these challenges, the Government of India implemented various policy measures and financial support packages to help sustain business and help recover the economy.

At the same time, India's international trade network was severely disrupted. The import dependence of different countries and commodities varied, leaving some more vulnerable to supply-chain shocks than others. Knowing how these disruptions affected trade performance and how to make the economy resilient to the next crisis was key for policymakers and businesses.

So resilience has become something that we need to study. Resilience is the ability of an economy, firm, sector or region to withstand shocks, adjust to new circumstances and recover from them. Examining resilience to COVID-19 reveals what makes businesses and economies resilient to large shocks.

1.2 Indian Context

India is one of the world's largest emerging economies and has extensive trade relations with several major economies. The country's industry and manufacturing sectors import large quantities of raw materials, energy products, machinery, chemicals and intermediate goods. Consequently, disruptions to international supply chains can have major implications for domestic production and performance.

During the COVID-19 pandemic, India's import demand experienced significant volatility due to changes in demand, logistics disruptions, and trade disruptions with key partners. Some groups of commodities, including mineral fuels, machinery, chemicals, and precious metals,

experienced fluctuations in vulnerability and recovery during the pandemic and post-pandemic periods.

Simultaneously, Indian MSMEs faced demand problems, labour replacement, liquidity problems and operational uncertainties. Government actions such as credit support programs and financial assistance schemes helped enterprises to survive and recover from pandemic-related disruptions.

, MSME resilience and international trade stability, and thus a full resilience assessment in these two areas is necessary to understand India's COVID-19 response and to provide lessons for future shocks.

1.3 Problem Statement

The COVID-19 pandemic laid bare weaknesses in domestic economies and global supply chains. While some firms, sectors and regions adapted well, others were long in recovery.

What is known is that resilience outcomes vary greatly across states, industries and trading relationships. However, there is little integrated analysis on how government support, labour market conditions, supply chain disruptions, country dependence, and trade exposure contributed to resilience during and after the pandemic.

Further, differences in import dependence among trading partners and commodity groups raise important questions about the sources of supply chain vulnerability and the determinants of trade recovery. Knowing these relationships is key to formulating policies to enhance economic resilience and reduce vulnerability to future shocks.

Therefore, this study explores resilience both domestically and internationally through MSME resilience, trade recovery, and COVID related supply chain exposure in India.

1.4 Research Questions

The study addresses the following research questions:

1. What are the key factors that influence MSME resilience in Indian states?

2. How does government support contribute to resilience?

3. Which trading partners and groups have contributed most to India's COVID-related supply-chain exposure?

4. How do supply shocks and country dependence affect trade performance?

5. What explains variation in COVID exposure by country and commodity?

6. Which sectors and countries recovered better after COVID?

7. What lessons can be learned to be more resilient against future economic crises?

1.5 Research Objectives

The study hopes to achieve the following objectives:

1. To assess the resilience performance of Indian states during and after the COVID-19 pandemic.

2. To study the impact of government support, employment and labour market indicators on MSME resilience.

3. To study the effect of COVID-19 on India's import supply chains with major trading partners and commodity groups.

4. To construct and evaluate a COVID Exposure measure of supply chain vulnerability during the pandemic.

5. To identify the major determinants of trade value and COVID exposure by econometric analysis.

6. To understand post-COVID trade recovery patterns between countries and sectors.

1.6 Scope of the Study

The study is on both domestic resilience and international trade resilience of the Indian economy.

MSME analysis uses state level indicators like resilience scores, government support measures, employment stress, labour participation, unemployment rates and MSME concentration. The trade analysis includes India's major import sectors and trading partners during the period of 2017–18 to 2024–25

For the trade component, five major HS commodity groups and thirteen key trading countries were chosen. The COVID period in the study includes 2020–21 and 2021–22 and post-COVID recovery period includes 2022–23 to 20

The study uses descriptive analysis, exploratory data analysis and regression modelling to examine resilience, trade recovery and supply chain vulnerability.

2. LITERATURE REVIEW

2.1 Literature on COVID-19 Spillovers, Supply Chain Disruptions and International Trade

The COVID-19 pandemic led to supply chain and trade disruptions worldwide. Several researchers have studied how these disruptions spread across countries through production linkages and trade dependencies. Bonadio et al. (2021) examined the role of global supply chains in transmitting COVID-19 shocks using a quantitative multi-country model and found that international production linkages accounted for a large Similarly, Baldwin and Freeman (2020) asserted that globalization increased the transmission of disruptions across economies, making countries dependent on a few suppliers more vulnerable. Ivanov and Dolgui (2020) investigated supply-chain resilience using simulation and emphasized the value of diversification, flexibility, and risk management. Miroudot (2020) looked at the resilience of trade during the pandemic and noted that countries with diversified import structures recovered faster than those with concentrated networks. Together they show that the vulnerability of a supply-chain is not just determined by the severity of disruption but also the structure of trade and dependence on key suppliers.

2.2 Literature on Firm Resilience, MSME Performance and Government Support During COVID-19

A huge body of research has been done on the resilience of firms and MSMEs during the pandemic. Bartik et al. (2020) surveyed small businesses in the United States and found severe financial distress, liquidity shortages, and operational disruptions in the early stages of the pandemic. Cowling et al. (2020) examined the impact of government assistance and found that it greatly improved business survival. Juergensen et al. (2020) studied policy measures in countries, finding that credit support, wage subsidies, and emergency financing had been essential in maintaining employment and business. In the Indian context, studies on ECLGS have found that access to credit improved liquidity and reduced business failure among MSMEs. Existing evidence thus suggests that institutional support, financial resilience and labour market stability are key factors in determining firm recovery outcomes during economic crises.

2. 3 Literature on Economic Recovery, Financial Interventions and Post-

After the initial disruption, there was more research into recovery dynamics across countries and sectors. Studies by the IMF and World Bank indicated that recovery patterns varied significantly depending on economic structure, policy interventions, sectoral composition and

external demand. Several studies have found that sectors related to manufacturing, industry, energy, technology and other mobility-related constraints have recovered more quickly than service sectors. Research on financial interventions also shows that credit expansion, fiscal stimulus programs and targeted assistance accelerated recovery by improving liquidity and business investment. The latest research suggests that resilience should not only be seen as being able to withstand shocks but as a dynamic process of resistance, adaptation and recovery. These findings emphasize the need to consider trade resilience, firm resilience, and policy support when measuring post-pandemic economic performance.

2. 4 Research Gap

Although extensive research has been done on COVID-19 disruptions, supply chain resilience, MSMEs performance and economic recovery, there are still many gaps. Most studies focus on firm level resilience or international trade vulnerability, but few study both. Existing research also typically analyses disruption and recovery separately, thereby missing the whole resilience process. Furthermore, little evidence on government support, labour market, supply chain vulnerability, COVID and post COVID recovery is available in Indian context. The present study addresses these gaps by developing an integrated framework to study MSME resilience, trade vulnerability, COVID exposure, sectoral credit support and post-pandemic recovery using several official datasets.

3. DATA SOURCES AND DATASET DESCRIPTION

3.1 Overview of Datasets Used

This study utilizes mainly 3 combined datasets

Table 3.1: Summary of dataset

Dataset	Purpose
Master MSME Dataset	Analysis of MSME resilience and economic stress
RBI Credit Dataset	Analysis of sector-wise financial support
Trade Dataset	Analysis of supply-chain disruptions and trade recovery

The Master MSME Dataset was created by integrating multiple government datasets, the RBI Credit Dataset and Trade Dataset were constructed separately.

3.2 Master Dataset

The Master Dataset was constructed by combining four datasets representing different Categories.

Datasets Integrated

MSME / Udyam Registration Dataset

Source:

- Ministry of MSME, Government of India

Purpose:

- Measures business concentration across states and districts.
- Represents the strength of the local MSME ecosystem.

Key Variables:

- State Name
- District Name
- Total Number of Enterprises
- MSME Category
- Registration Date

ECLGS Loan Dataset

Source:

- Emergency Credit Line Guarantee Scheme (ECLGS)

Purpose:

- Measures government financial assistance provided during COVID-19.
- Acts as a Government Support Indicator.

Key Variables:

- State/UT
- Number of Guarantees
- Loan Amount
- Guarantee Amount

COVID Cases Dataset

Source:

- State-wise COVID Statistics

Purpose:

- Measures pandemic severity across states.

Key Variables:

- State/UT
- Total COVID Cases

Unemployment Dataset

Source:

- Labour Market Statistics

Purpose:

- Measures economic stress during COVID-19.

Key Variables:

- State
- Unemployment Rate
- Labour Participation Rate
- Month

3.3 Construction of the Master Dataset

The four datasets were merged at the state level to create a unified dataset for resilience analysis.

The integrated dataset contains information on:

- Number of covid cases per state
- Unemployment in India on different years
- ECLGS loan details
- Concentration of MSME across states

This dataset forms the basis for the resilience analysis conducted in the study.

3.4 Resilience Score Construction

To evaluate relative resilience across states, a Resilience Score was constructed using employment-related indicators.

The score was calculated as:

Equation 3.1:

$$Resilience\ Score = \frac{Estimated\ Employed}{Maximum\ Estimated\ Employed}$$

If the result is 1, it indicates stronger resilience.

If the result is 0, it indicates weaker resilience.

3.5 RBI Credit Dataset

Source

Reserve Bank of India (RBI)

Dataset Name

RBI Sectoral Credit Deployment Dataset

Purpose

This dataset contains the amount of credit provided to different sectors in the business during the COVID-19 period.

Businesses depend on credit for:

- Working Capital
- Salary Payments
- Inventory Purchases

- Business Expansion
- Operational Continuity

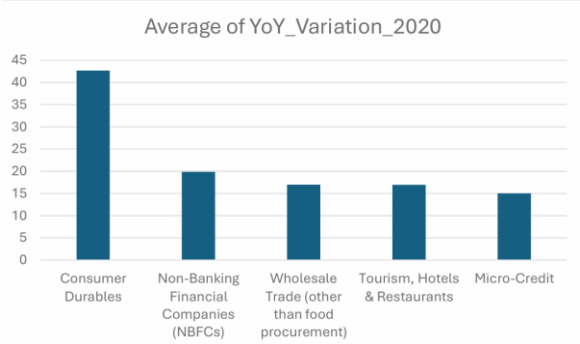
Key Variables

- Sector
- Credit Outstanding
- Year-on-Year Variation
- Financial Year Variation
- Month

Analytical Use

The dataset is used to evaluate sector-wise financial support and identify sectors who are receiving significant credit expansion during the pandemic.

Figure 3.1: RBI credit summary



3.6 Trade Dataset

Source

- TradeStat portal

Study Period

Pre-COVID Period

- 2017–18
- 2018–19
- 2019–20

COVID Period

- 2020–21
- 2021–22

Post-COVID Period

- 2022–23
- 2023–24
- 2024–25

Unit of Analysis

- Country
- HS Commodity Group
- Year

Trade values are measured in Crores

3.7 Selection of Major HS Commodity Groups

The main task is to find the best HS commodity groups which contributes to India's import, the most important top five HS chapters are selected from the trade stat portal which take more than 50% share in the total import of India. This HS codes are selected for detailed analysis.

Table 3.2: Major HS Commodity Groups Selected

HS Code	Commodity Group
HS27	Mineral Fuels and Mineral Oils
HS29	Organic Chemicals
HS71	Precious Stones and Metals
HS83	Miscellaneous Articles of Base Metal
HS84	Machinery and Mechanical Appliances

Table 3.3: Import Share of Selected HS Commodity Groups

HS Code	Commodity Group	Import Value (₹ Cr)	Share (%)
HS27	Mineral Fuels, Mineral Oils & Products	1,36,91,354	55.58
HS71	Precious Stones & Metals	53,88,277	21.87
HS84	Machinery & Mechanical Appliances	36,48,277	14.81
HS29	Organic Chemicals	18,26,444	7.41
HS83	Miscellaneous Articles of Base Metal	79,123	0.32
Total		2,46,33,475	99.99

Figure 3.2: Distribution of Import Share among Selected HS Commodity Groups

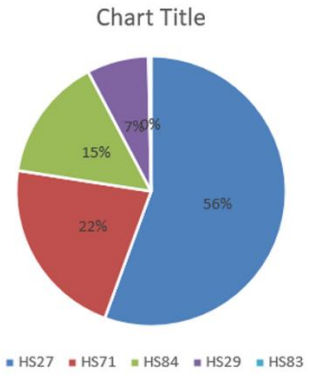
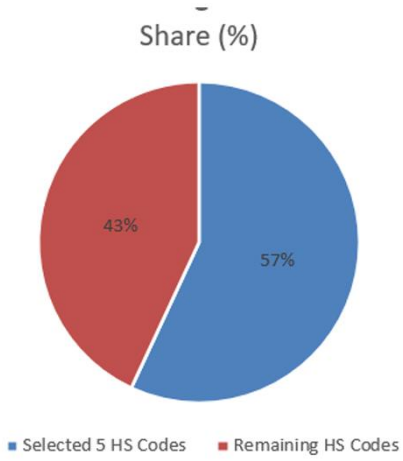


Figure 3.3: Share of Selected HS Commodity Groups versus Remaining HS Chapters



Observation:

These five HS chapters collectively account for approximately 57% of the total imports considered in this study. Among that 5 HS chapters, HS27 alone contributes 55.58%, this indicates that India is highly depended on energy products such as crude oil and petroleum products.

3.8 Selection of Major Trading Partners

After identifying the major commodity groups, the next major task is to select the most important countries where India is depending on import.

Countries were ranked according to their total contribution to India's import value. The top importing partners, along with strategically important suppliers for the selected HS chapters, were retained for detailed resilience assessment.

A total of thirteen countries were selected because together they account for nearly two-thirds of India's total imports, making them the most critical contributors to supply chain dependency.

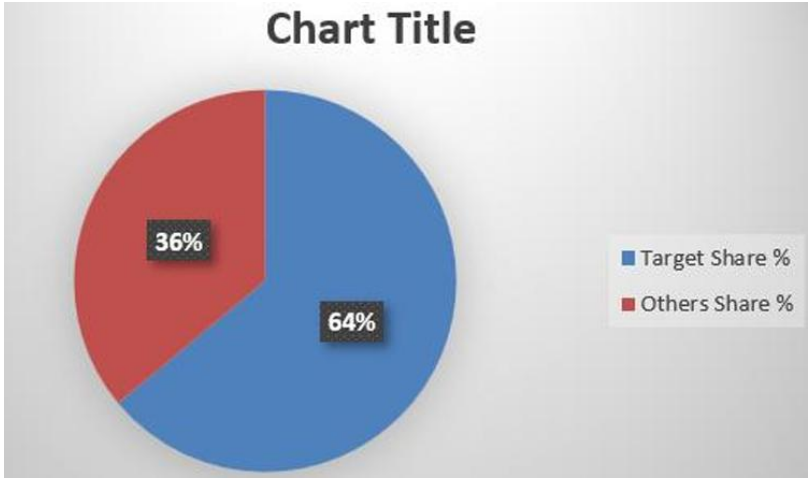
Selected Countries

- Belgium
- China
- Germany
- United States
- Japan
- South Korea
- Singapore
- UAE
- Saudi Arabia
- Iraq
- Switzerland
- United Kingdom
- Netherlands

Table 3.4: Import Contribution of Selected Countries

Metric	Value
Imports from Selected 13 Countries	₹ 1,57,57,462.68 Cr
Imports from Remaining Countries	₹ 88,76,012.81 Cr
Grand Total Imports	₹ 2,46,33,475.49 Cr
Share of Selected Countries	63.97%
Share of Remaining Countries	36.03%

Figure 3.4: Share of Selected Countries in Total Imports



Observation:

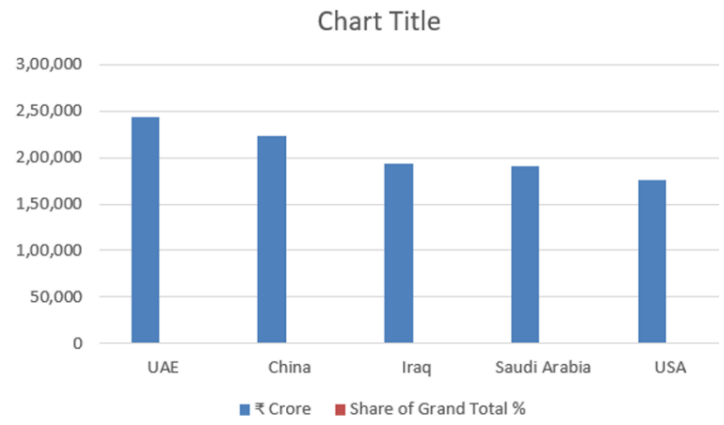
The selected thirteen countries contribute approximately 63.97% of India's total imports. This confirms that a relatively small group of trading partners dominates India's import structure.

Any disruption affecting these countries can significantly influence India's supply chains, making them highly relevant for resilience analysis.

Table 3.5: Top Five Contributors among Selected Countries

Rank	Country	Import Value (₹ Cr)	Share of Total Imports (%)
1	UAE	2,43,077	9.87
2	China	2,23,269	9.06
3	Iraq	1,93,212	7.84
4	Saudi Arabia	1,90,807	7.75
5	USA	1,75,227	7.11
Top 5 Total		10,25,592	41.63

Figure 3.5: Top Five Import-Contributing Countries



Observation:

The UAE emerges as India's largest import partner, contributing 9.87% of total imports, the next one is China at 9.06%.

The top five countries contribute 41.63% of India's total imports.

3.9 Variable Definitions

Table 3.3: Variable definitions

Variable	**Description**
Resilience Score	State-level resilience indicator
Support per MSME	Government support per enterprise
Employment Stress	Employment-related pressure indicator

Unemployment Rate	State-level unemployment rate
Labour Participation Rate	Workforce participation rate
Trade Value	Import value in USD Million
Supply Shock	Deviation from pre-COVID imports
Country Weight	Relative importance of supplier country
Weighted Shock	Weighted supply disruption
COVID Exposure	Supply-chain vulnerability indicator

4. RESEARCH METHODOLOGY

4.1 Research Framework

The study is made on all 3 datasets separately for pre covid, covid and post covid periods, The main intakes are:

1. MSME Resilience Analysis
2. Trade Value Analysis
3. COVID Exposure Analysis

At the national level, resilience is affected by government support, labour market conditions, employment stress, and MSME concentration. At the international level, trade performance and supply-chain vulnerability are affected by supply shocks, dependence on trading partners, and commodity-specific characteristics.

4.2 Research Design

A quantitative research design is done on secondary data collected from official government and institutional sources.

The analytical process consists of:

1. Data Collection
2. Data Cleaning and Preparation
3. Construction of COVID Exposure Measure
4. Exploratory Data Analysis (EDA)
5. Descriptive Statistical Analysis
6. Correlation Analysis
7. Regression Modelling
8. Interpretation of Results

4.3 Data Preparation and Cleaning

Before analysis, data cleaning is done to ensure reliability and consistency in the dataset and to prevent from false predictions.

The following steps were performed:

- Removal of duplicate observations
- Treatment of missing values
- Verification of country and HS code classifications

- Standardization of variable formats
- Validation of calculated variables
- Identification of extreme observations and outliers

For the trade dataset, observations with missing values in important variables such as Import Growth, Supply Shock, Country Weight, and COVID Exposure were removed from the analysis.

4.4 Construction of COVID Exposure Measure

4.4.1 Pre-COVID Average Imports

The pre-pandemic baseline was established using import values from:

- 2017–18
- 2018–19
- 2019–20

Equation 4.1

$$\text{Pre-COVID Average} = \frac{\text{Import}_{2017-18} + \text{Import}_{2018-19} + \text{Import}_{2019-20}}{3}$$

4.4.2 Supply Shock

Supply Shock measures the percentage deviation of COVID-period imports from the pre-pandemic benchmark.

Equation 4.2

$$Supply\ Shock = \left(\frac{Current\ Import - PreCOVID\ Average}{PreCOVID\ Average}\right) \times 100$$

Observation:

- Positive values indicate import expansion.
- Negative values indicate import contraction.

4.4.3 Country Weight

Country Weight measures the relative importance of a supplier country.

Equation 4.3

$$Country\ Weight = \frac{Import_{i,k}}{\sum_{i=1}^{n} I\,mport_{i,k}}$$

4.4.4 Weighted Shock

Equation 4.4

$$Weighted\ Supply\ Shock = Supply\ Shock \times Country\ Weight$$

4.4.5 COVID Exposure

Equation 4.5

$$COVID\ Exposure = \sum_{i=1}^{n} (Weighted\ Supply\ Shock_i)$$

The measure captures overall supply-chain vulnerability experienced during the pandemic.

4.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted before regression modelling to understand dataset characteristics and identify patterns.

MSME Analysis

- Top Resilient States
- Least Resilient States
- COVID Burden Analysis
- Government Support Analysis

Trade Analysis

- Country-wise COVID Exposure
- HS Commodity Analysis
- Import Growth Analysis
- Supply Shock Analysis
- Post-COVID Recovery Analysis

Techniques used:

- Summary Statistics
- Histograms
- Bar Charts
- Scatter Plots
- Heatmaps

4.6 Descriptive Statistical Analysis

Descriptive statistics were calculated to summarize the characteristics of the datasets.

Measures calculated include:

- Mean
- Median
- Standard Deviation
- Minimum
- Maximum
- Range
- Skewness
- Kurtosis

These measures provide insights into the distribution and variability of key variables.

4.7 Correlation Analysis

Correlation analysis was conducted using the Pearson Correlation Coefficient to examine relationships among variables.

Variables analysed include:

- Import Growth
- Supply Shock
- Country Weight
- Weighted Shock
- COVID Exposure
- MSME Resilience Variables

Equation 4.6

$r = Cov(X,Y) \div (\sigma X \times \sigma Y)$

where:

- r = Correlation Coefficient
- Cov(X,Y) = Covariance
- σX = Standard Deviation of X
- σY = Standard Deviation of Y

4.8 Regression Methodology

To identify the determinants of resilience, trade value, and COVID exposure, multiple linear regression models were estimated.

Equation 4.7

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n + \varepsilon$$

where:

- Y = Dependent Variable
- β_0 = Intercept
- β_i = Regression Coefficients
- X_i = Independent Variables
- ε = Error Term

4.9 MSME Resilience Models

Dependent Variable

- Resilience Score

Independent Variables

- Support per MSME
- Employment Stress
- Unemployment Rate
- Labour Participation Rate
- MSME Population

Model 1A

Resilience Score = f (Support per MSME)

Model 1B

Resilience Score = f (Support per MSME, Employment Stress)

Model 1C

Resilience Score = f (Support per MSME, Employment Stress, Unemployment Rate)

Model 1D

Resilience Score = f (Support per MSME, Employment Stress, Unemployment Rate, Labour Participation Rate)

Model 1E

Resilience Score = f (Support per MSME, Employment Stress, Unemployment Rate, Labour Participation Rate, MSME Population)

4.10 Trade Value Models

Dependent Variable

- Trade_Value_USD_Mn

Independent Variables

- COVID Exposure
- Supply Shock
- Country Weight
- Commodity Growth
- Country Growth
- Commodity Share

Model 2A

Trade Value= f (COVID Exposure)

Model 2B

Trade Value = f (COVID Exposure, Supply Shock)

Model 2C

Trade Value = f (COVID Exposure, Supply Shock, Country Weight)

Model 2D

Trade Value = f (COVID Exposure, Supply Shock, Country Weight, Commodity Growth, Country Growth)

Model 2E

Trade Value = f (COVID Exposure, Supply Shock, Country Weight, Commodity Growth, Country Growth, Commodity Share)

4.11 COVID Exposure Models

Dependent Variable

- COVID Exposure

Independent Variables

- Supply Shock
- Country Weight
- Commodity Growth
- Country Growth
- Commodity Share

Model 3A

COVID Exposure = f (Supply Shock)

Model 3B

COVID Exposure = f (Supply Shock, Country Weight)

Model 3C

COVID Exposure = f (Supply Shock, Country Weight, Commodity Growth)

Model 3D

COVID Exposure = f (Supply Shock, Country Weight, Commodity Growth, Country Growth)

Model 3E

COVID Exposure = f (Supply Shock, Country Weight, Commodity Growth, Country Growth, Commodity Share)

4.12 Software and Tools Used

The analysis was conducted using:

- Microsoft Excel
- Python
- Statistical Regression Tools
- Data Visualization Libraries
- AI-Assisted Analytical Workflows

5. RESULTS AND FINDINGS

5.1 MSME Resilience Analysis

The MSME dataset was analysed to evaluate resilience across Indian states using the constructed Resilience Score.

5.1.1 Distribution of Resilience Scores

The average resilience score was 0.247, while the median value was 0.160. The distribution was positively no straight, this indicates that a very small number of states had high resilience.

Table 5.1: Summary Statistics of Resilience Scores

Statistic	Value
Mean	0.2469
Median	0.1604
Standard Deviation	0.2387
Minimum	0.0033
Maximum	1.0000
Range	0.9967
Skewness	1.3009
Kurtosis	1.2766
Observations	262

Observation:

This result shows that large variation in resilience across states. The gap between the highest and lowest scores indicates unequal recovery capabilities .

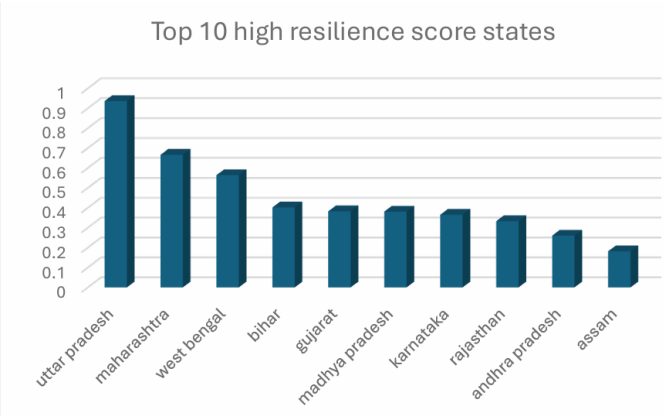
5.1.2 Top Resilient States

Table 5.2: Top 10 Most Resilient States

Rank	State	Average Resilience Score
1	Uttar Pradesh	0.936
2	Maharashtra	0.666
3	West Bengal	0.564
4	Bihar	0.402
5	Gujarat	0.383

6	Madhya Pradesh	0.381
7	Karnataka	0.365
8	Rajasthan	0.333
9	Andhra Pradesh	0.260
10	Assam	0.182

Figure 5.1: Top 10 Most Resilient States



Observation:

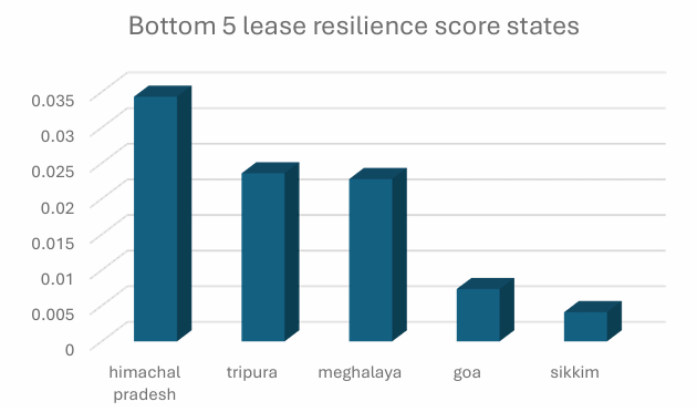
Uttar Pradesh has the highest resilience score, second one is Maharashtra and next West Bengal. These states showed stronger recovery performance and greater capacity to withstand pandemic-related disruptions.

5.1.3 Least Resilient States

Table 5.3: Bottom 5 Least Resilient States

Rank	**State**	**Average Resilience Score**
1	Sikkim	0.004
2	Goa	0.007
3	Meghalaya	0.023
4	Tripura	0.024
5	Himachal Pradesh	0.034

Figure 5.2: Least Resilient States



Observation:

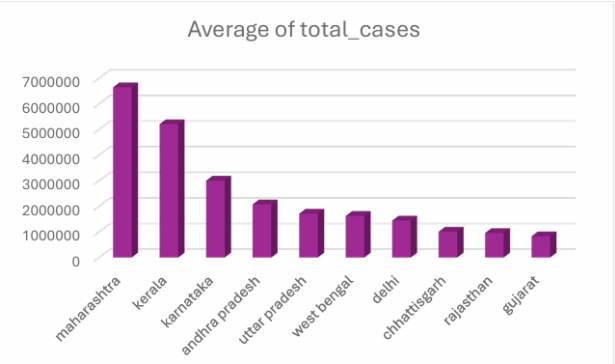
States like Sikkim, Goa, Meghalaya, Tripura, and Himachal Pradesh has comparatively lower resilience scores, this shows weaker recovery performance.

5.1.4 COVID Burden Analysis

Table 5.4: States with Highest COVID Burden

Rank	State	Average COVID Cases
1	Maharashtra	6,646,938
2	Kerala	5,207,826
3	Karnataka	3,001,554
4	Andhra Pradesh	2,075,419
5	Uttar Pradesh	1,710,617
6	West Bengal	1,625,375
7	Delhi	1,441,935
8	Chhattisgarh	1,007,327
9	Rajasthan	955,173
10	Gujarat	828,367

Figure 5.3: Top States by COVID Cases



Observation:

Maharashtra, Kerala, and Karnataka had the highest COVID-19 cases. Several highly affected states also showed strong resilience, indicating that resilience depends on both disruption severity and recovery capacity.

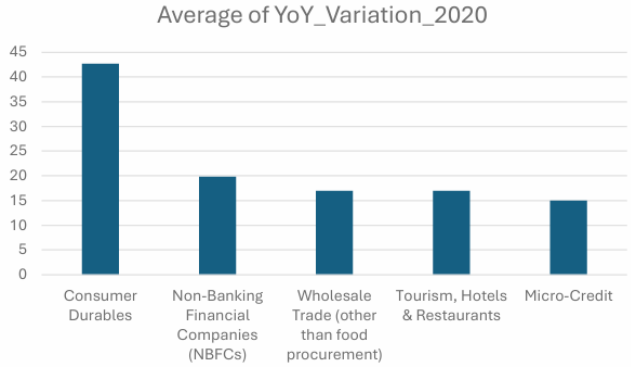
5.2 RBI Credit Analysis

RBI details were analysed to check financial support during pandemic and also to check secyor wise credit growth.

Table 5.5: Top Five Sectors by Credit Growth During COVID-19

Rank	**Sector**	**Average YoY Credit Growth (%)**
1	Consumer Durables	42.63
2	Non-Banking Financial Companies (NBFCs)	19.84
3	Wholesale Trade (Other than Food Procurement)	16.96
4	Tourism, Hotels & Restaurants	16.93
5	Micro-Credit	14.99

Figure 5.4: Sector-wise Credit Growth During COVID-19



Observation:

Consumer Durables recorded the highest average credit growth, followed by NBFCs, Wholesale Trade, Tourism & Hospitality, and Micro-Credit sectors. The findings indicate that credit expansion played an important role in supporting economic activity during the pandemic.

5.3 COVID Period Trade Analysis

5.3.1 Dataset Overview

Table 5.6: COVID Trade Dataset Overview

Attribute	Description
Study Period	2020–21 to 2021–22
Number of Trading Partners	13
Number of HS Commodity Groups	5
Unit of Analysis	Country–HS Code–Year
Trade Measure	Import Value (USD Million)
Variables Included	Trade Value, Import Growth, Supply Shock, Country Weight, Weighted Shock, COVID Exposure
Data Source	DGCI&S TradeStat / Ministry of Commerce & Industry
Purpose	Analysis of trade disruptions and supply-chain vulnerability during COVID-19

5.3.2 Country-wise COVID Exposure Analysis

Table 5.7: Average COVID Exposure by Country

Rank	Country	Average COVID Exposure
1	China	16.181
2	UAE	6.986
3	USA	4.445
4	Switzerland	3.027
5	Iraq	2.988
6	Singapore	2.858
7	Hong Kong	2.166
8	Saudi Arabia	0.666

9	Russia	0.505
10	Belgium	0.416
11	Korea	-0.060
12	Japan	-0.109
13	Germany	-0.669

Observation:

China recorded the highest COVID Exposure among all trading partners, this says India was highly dependent on Chinese supply chains. UAE and the United States also showed relatively high exposure levels.

5.3.3 HS Commodity Analysis

Table 5.8: Average Import Growth and Supply Shock by HS Code

HS Code	**Commodity Group**	**Avg. Import Growth (%)**	**Avg. Supply Shock (%)**
HS27	Mineral Fuels & Oils	53.19	46.02
HS29	Organic Chemicals	21.20	21.75
HS84	Machinery & Mechanical Appliances	19.16	20.93
HS83	Articles of Base Metal	15.24	12.76
HS71	Precious Stones & Metals	1327.70	10.00

Observation:

HS27 (Mineral Fuels and Oils) experienced the largest supply shock during the pandemic, and HS71 (Precious Stones and Metals) exhibited strong recovery.

5.4 Post-COVID Trade Recovery Analysis

5.4.1 Country Recovery Analysis

Table 5.9: Highest Import Recovery by Country

Rank	**Country**	**Average Import Growth (%)**
1	Hong Kong	84.32
2	Russia	56.93
3	Iraq	31.33
4	Saudi Arabia	28.01

5	Korea	24.21

Observation:

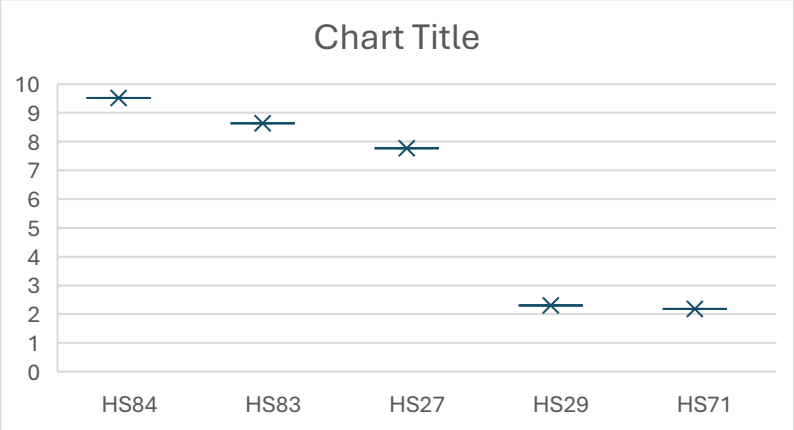
Hong Kong, Russia, Iraq, and Saudi Arabia recorded the strongest post-pandemic recovery in import performance.

5.4.2 Sector Recovery Analysis

Table 5.10: Sector-wise Recovery Performance

HS Code	**Commodity Group**	**Median Import Growth (%)**	**Median Supply Shock (%)**
HS27	Mineral Fuels & Oils	7.77	133.03
HS29	Organic Chemicals	2.30	42.56
HS71	Precious Stones & Metals	2.18	17.64
HS83	Articles of Base Metal	8.64	47.49
HS84	Machinery & Mechanical Equipment	9.52	43.71

Figure 5.5: Recovery Across Commodity Groups

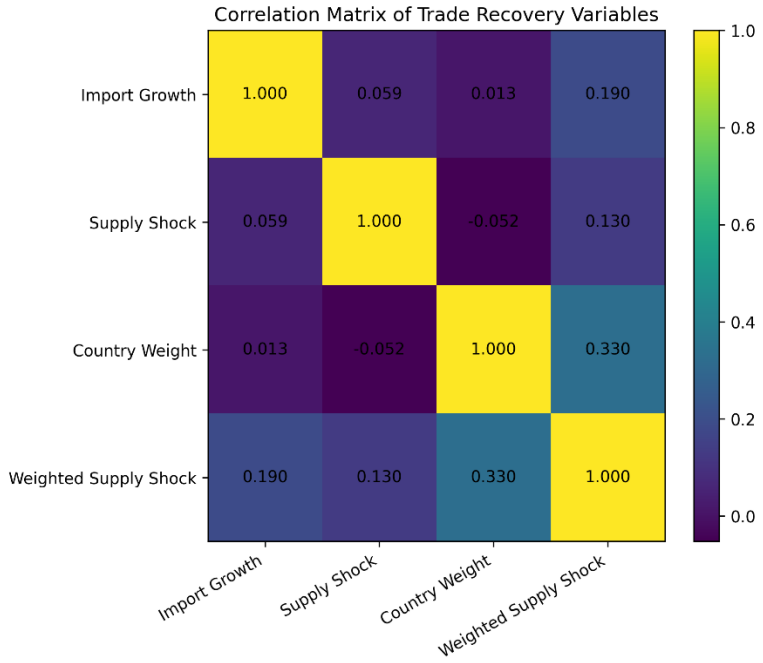


Observation:

HS84 (Machinery and Mechanical Equipment) showed the strongest recovery performance, whereas HS27 remained highly vulnerable to external disruptions.

5.5 Correlation Analysis

Table 5.6: Correlation Matrix



Observation:

Key observations include:

- Weak relationship between Import Growth and Supply Shock.
- Weak relationship between Import Growth and COVID Exposure.
- Strong positive relationship between Country Weight and COVID Exposure.
- Strong positive relationship between Weighted Shock and COVID Exposure.

5.6 Regression Analysis

5.6.1 MSME Resilience Models

Table 5.12: Regression Results – MSME Resilience Models (1A–1E)

Variables	**Model 1A**	**Model 1B**	**Model 1C**	**Model 1D**	**Model 1E**
Support per MSME	-3.632***	-3.591***	-3.694***	-3.694***	-1.144***
Employment Stress	—	0.002	0.002	0.000	-0.0001
Unemployment Rate	—	—	-0.0064***	-0.0064***	-0.0040***
Labour Participation Rate	—	—	—	0.000	-0.0001

MSME Population	—	—	—	—	3.16E-07***
Constant	0.470***	0.342***	0.436***	0.631***	0.244***
Observations	262	262	262	262	262
R^2	0.348	0.353	0.416	0.416	0.638
Adjusted R^2	0.346	0.348	0.405	0.405	0.629
F Statistic	138.80	70.55	61.30	61.30	113.40

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Observation:

Government support per MSME emerged as the most important determinant of resilience. Labour market indicators, particularly unemployment rate, significantly improved model performance.

5.6.2 Trade Value Models

Table 5.13: Regression Results – Trade Value Models (2A–2E)

Variables	**Model 2A**	**Model 2B**	**Model 2C**	**Model 2D**
COVID Exposure	3067.35***	3050.91***	2443.55***	1449.31***
Supply Shock	—	7.83	604.96	100.94*
Country Weight	—	—	2413.56***	2443.55***
Commodity % Growth	—	—	—	30.95
Country % Growth	—	—	—	100.94*
Constant	12604.46***	12476.86***	6243.56***	-4135.37
Observations	124	124	124	124
R^2	0.302	0.303	0.368	0.860
Adjusted R^2	0.297	0.291	0.352	0.852
F Statistic	52.90	26.25	23.31	102.14

Observation:

COVID Exposure explained a substantial portion of variation in trade value. The addition of Country Weight significantly improved explanatory power, indicating the importance of supplier-country dependence.

5.6.3 COVID Exposure Models

Table 5.14: Regression Results – COVID Exposure Models (3A–3E)

Model	Independent Variables	R^2	Adjusted R^2	Conclusion
3A	Supply_Shock	0.069	0.062	Weak explanatory power
3B	+ Country_Weight	0.672	0.667	Major improvement
3C	+ Commodity_%_Growth	0.681	0.673	Small improvement
3D	+ Country_%_Growth	0.681	0.671	Very little improvement
3E	+ %_Share_of_Commodity	0.682	0.669	Best model numerically

Observation:

Country Weight is strongest indicator of COVID Exposure. The findings shows that vulnerability is depending on major trading partners rather than disruption intensity alone.

5.7 Major Findings

1. Resilience varied differently in each state.
2. The support provided by Government helped to improve MSME resilience.
3. Labour market conditions mainly affected resilience outcomes.
4. Credit expansion supported economic stability.
5. China was the highest COVID Exposure among India's trading partners.
6. HS27 experienced the largest supply-chain disruption.
7. Post-COVID recovery differed significantly across countries and sectors.

6. MANAGERIAL INSIGHTS

6.1 Insights on MSME Resilience and Financial Support

This analysis tells that resilience is different across Indian states. States with higher government support generally has good resilience scores and recovered faster from the pandemic. Employment conditions also played an important role in recovery. The RBI credit analysis further showed that sectors such as Consumer Durables, NBFCs, Wholesale Trade, Tourism & Hospitality, and Micro-Credit received higher credit support during the pandemic. This indicates that both government assistance and financial support helped businesses manage disruptions and improve recovery.

Key Insights

- The support provided by Government helped to improve MSME resilience.
- Employment conditions affected recovary.
- Credit expansion helped businesses to continue work during the pandemic.
- Financial support is the main factor during economic unstability.

6.2 Insights on Trade Vulnerability and Recovery

The trade analysis showed significant differences in disruption and recovery across countries and commodity groups. China has the highest COVID Exposure, indicating a high level of dependence on Chinese supply chains. HS27 experienced the largest supply shock, while HS84 showed strong recovery after the pandemic. Countries such as Hong Kong, Russia, Iraq, and Saudi Arabia also demonstrated strong recovery performance.

Key Insights

- Dependence on a few supplier countries increases vulnerability.
- Supply-chain disruptions affected sectors differently.
- Recovery patterns varied across countries and commodity groups.
- Diversification of suppliers can reduce future risks.

6.3 Policy Implications and Study Contributions

The regression results highlighted the importance of government support, Country Weight, and COVID Exposure in explaining resilience and trade performance. The study combines MSME resilience and trade resilience within a single framework and introduces a COVID Exposure measure to assess supply-chain vulnerability.

Key Insights

- Continue targeted support for MSMEs during crises.
- Improve employment opportunities and workforce participation.

- Strengthen trade partnerships with multiple countries.
- Promote data-driven decision making for risk management.
- Reduce dependence on concentrated supply chains.

7. CONCLUSION

Major Findings

- This is a analysis of MSME resilience, RBI credit growth, trade disruptions, and supply-chain vulnerability during and after COVID-19 in India.
- Resilience is different in different states, with states receiving higher government support generally showing better recovery.
- Labour market conditions such as employment and workforce participation also influenced resilience outcomes.
- RBI credit analysis showed strong credit growth in sectors such as Consumer Durables, NBFCs, Wholesale Trade, Tourism & Hospitality, and Micro-Credit.
- China recorded the highest COVID Exposure among India's major trading partners, indicating high dependence on Chinese imports.
- HS27 (Mineral Fuels and Oils) experienced the highest supply shock during the pandemic period.
- HS84 (Machinery and Mechanical Equipment) showed stronger recovery during the post-COVID period.
- Regression results showed that government support was an important factor affecting MSME resilience.
- Country Weight emerged as the most significant factor influencing trade vulnerability and COVID Exposure.

Concluding Remarks

- Financial support and government interventions helped businesses manage pandemic-related disruptions.
- Stable employment conditions contributed to stronger resilience and recovery.
- Dependence on a limited number of supplier countries increased supply-chain risk.
- Diversified trade relationships can help reduce future vulnerabilities.
- Resilience depends not only on resisting shocks but also on the ability to adapt and recover.
- The findings can support policymakers and researchers in developing strategies for future economic disruptions.

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